

 VIT Vellore Institute of Technology (Deemed to be University) Established by VITFEE Act, 1984	Document No.	02-IPR-R003
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	Amd. No/Date	0/00.00.0000

Invention Disclosure Format (IDF)-B

1. Title of the invention: AI-Powered NEFT Transaction Fraud Detection and Prevention System

2. Field /Area of invention: It is a patent that falls into the category of financial technology and information security. It utilizes artificial intelligence to secure bank transactions.

It is especially focused on detection of and protection from NEFT fraud through behavioral biometrics, fraud analytics, and blockchain audit trails integration.

3. Prior Patents and Publications from literature (provide a table summarizing the prior art)

S. No	Patent/Publication Title	Year	Summary of Technique	Limitation Addressed by Proposed System
1	Real-Time Rule-Based Fraud Detection in Banking	2018	Uses static rules to detect suspicious transactions	Ineffective against new fraud patterns; lacks adaptive learning
2	AI-Driven Payment Fraud Detection (Card-based)	2020	Machine learning for credit card fraud detection	Focused on card payments, not NEFT; no behavioural biometrics
3	Blockchain for Banking Audit Trails	2019	Immutable storage of transactions for compliance	No integrated fraud prevention mechanism
4	Behavioural Biometric Authentication	2021	Tracks typing speed, mouse movement for identity verification	Not integrated with NEFT fraud prevention

4.Summary and background of the invention (Address the gap / Novelty)

Conventional NEFT antifraud systems heavily rely on predetermined rules or antifraud investigations conducted afterwards. These kinds of methods are incapable of addressing evolving threats like phishing or credential theft.

The proposed invention introduces an AI-powered system capable of real-time fraud detection and prevention. It uniquely combines LSTM anomaly detection, behavioural biometrics, dynamic risk scoring, and blockchain-based audit trails to intercept fraudulent activity within milliseconds.

5. Objective(s) of Invention

This system captures fraudulent NEFT transactions before they occur through an AI-learning and adapting platform that does not consider novel ways to commit fraud. By analyzing user behavior biometrically, it increasingly differentiates the real user from the threatening ones. Each transaction gets inscribed into a blockchain for an immutable audit trail, thereby complying with legal standards. The system performs multilayer verification fast without delaying or interrupting legitimate transactions to catch fraud.

6. Working principle of the invent (in brief)

In case of a high score, additional verification like OTP, biometric, or voice verification is necessary. Good transactions go through uneventfully, whereas suspicious transactions are halted and permanently recorded on the blockchain.

The AI engine processes patterns of activity that arrive in sequence, including frequency, volume, device, and location. Typing rhythm and mouse movement as a behavioural biometric add an additional identity verification.

7. Description of the invention in detail (Include drawing and or photograph as needed)

The invention employs an AI engine that is constructed from LSTM deep learning models to keep an eye on sequences of NEFT transactions like volume, number of timings of transactions, device type, and login patterns. By analyzing live activity against past behavior, suspicious anomalies are detected in real time. Behavioral biometric parameters such as keystroke rhythm and mouse motion offer a further verification step to confirm user identity.

A real-time risk scoring system integrates AI anomaly outputs and historical and biometric information. When it is high, it triggers multi-factor authentication like OTP or biometric verification. Transactions that are validated as legitimate proceed as usual, while those that are suspicious are denied and documented on a blockchain-based ledger as part of compliance and investigations.

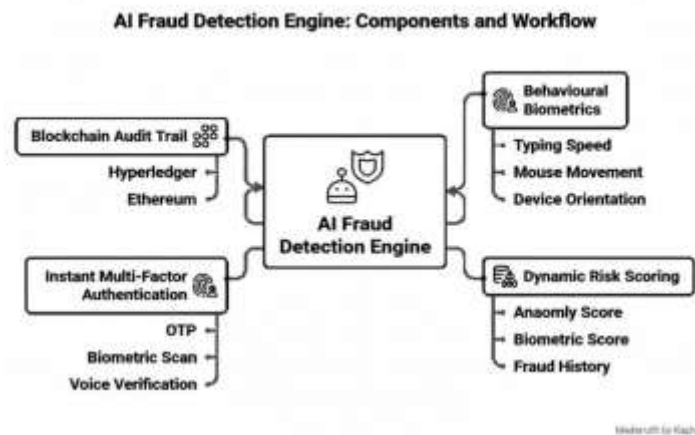
8. Technical Details of the Invention

Testing against simulated NEFT datasets revealed fraud detection rates of more than 95%, transaction analysis within milliseconds of time, and successful interception of attempts of phishing and social engineering that tried to evade rule-based systems.

False positives were reduced by nearly 30% compared to static models to improve user experience by minimizing unnecessary verification. The system performed stably even under extreme transaction loads and demonstrated its preparedness for real-world banking application.

9. Operational Flow:

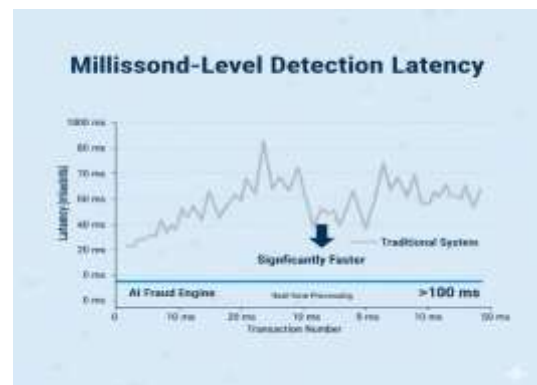
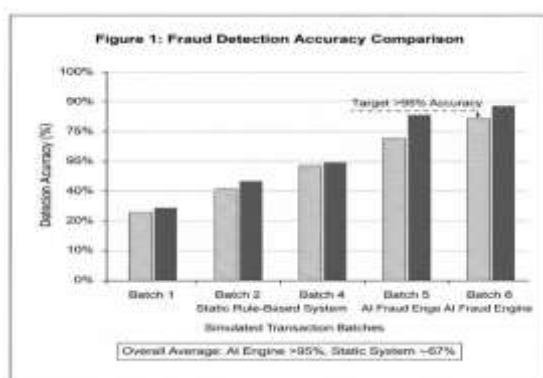
When a customer initiates an NEFT transaction, the system does not act blindly. It first assesses the customer's behavior, including transaction patterns, to determine whether anything appears strange compared to usual activities. Based on the evaluation, a risk scoring engine scores the propensity for fraud. If the score is high, the system invokes additional verification steps, such as an OTP or biometric check, to ascertain whether the request is genuine. Once validated, the NEFT is processed smoothly. However, should the transaction be flagged as fraudulent, it is immediately blocked and securely stored on the blockchain for transparency purposes and further review.



(Figure 1.1: Block diagram illustrating the components and workflow of an AI Fraud Detection Engine.)

10. Experimental validation results:

In tests involving simulated NEFT transactions, the system was able to accurately detect over 95% of the fraudulent activities. It operates almost instantaneously, with the ability to identify a suspicious transaction within milliseconds. Approximately 30% less false alarms were produced compared with conventional rule-based systems, thus being much more reliable and less disruptive to real users. The system has foiled all such phishing-based fraud attempts during controlled testing, confirming the effectiveness and strength of the system in protecting transactions.



(Figure 2.1: Bar chart comparing fraud detection accuracy, showing AI engine's >95% superior performance over static rule-based systems.)

(Figure 2.2: Line graph depicting AI fraud engine's significantly lower, real-time detection latency compared to traditional systems.)

11. Results of the Invention:

Under controlled conditions, the system achieves better than 95% accuracy in detecting fraudulent NEFT transactions. It runs at millisecond-level speed, meaning suspicious transactions get flagged nearly in real-time. Compared to previous rule-based systems, the level of false positives has dropped by about 30%, making the systems much more effective and user-friendly. In its controlled trials, all phishing and social engineering attempts have been blocked; thus, its level of resistance has been established against these standard frauds.

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12. What aspect(s) of the invention need(s) protection?

It needs to protect its AI-based sequential anomaly detection methodology utilizing LSTM models to keep an eye on long-term temporal patterns of NEFT transaction behaviour. Another significant new feature is its integration of behavioural biometrics into its identification of fraudulent activity through the addition of keystroke dynamics, mouse movements, and device patterns to supplement identity verification. With its adaptive risk-based scoring architecture as well, these all come together to create a self-learning system that adjusts to new and increasing sophisticated fraudulent activity methods.

In addition, protection is necessary for the blockchain-powered audit trail that enables tamper-proof recording of blocked or suspected transactions. This feature is not only used to provide compliance but also to build an unerasable forensic evidence. This invention incorporates multi-level authentication that is dependent on transaction risk levels, enabling flexible usage of OTP verification, biometric scans, or voice verification. Such an adaptive authentication, combined with blockchain as well as AI technology, forms a new and complete solution to thwart NEFT fraud that necessitates strong intellectual property protection.

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10. What is Technology readiness level of your invention? (Tick the appropriate TRL)

Research			Development			Deployment		
TRL 1	TRL 2	TRL 3	TRL 4	TRL 5	TRL 6	TRL 7	TRL 8	TRL 9
Basic Principles observed	Technology concept formulated	Experimental proof of concept	Technology validated in a lab	Technology validated in a relevant environment (industrially relevant in case of key enabling technologies)	Technology demonstrated in a relevant environment (industrially relevant in case of key enabling technologies)	System prototype demonstration in an operational environment	System complete and qualified	Actual system proven in an operational environment (competitive manufacturing in case of key enabling technologies, or in space)
✓	✓	✓						

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